SASI KANDURI

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WORK EXPERIENCE

Office of Water Programs — **Sacramento, CA**
Software Engineer, Software Engineer Intern — **Jun 2023 – Present**

- Led the development of a Water Quality Testing System for the State Water Resources Control Board of California using **React, Flask, SQLAlchemy, Jinja, Redis** and **MS SQL** to enable sampling, laboratory testing, and results for PFAS chemicals across 4,000 wells.
- Designed and implemented a Stormwater Analytics tool for the California Department of Transportation, allowing state consultants to report real-time storm data, analyze contamination, and predict water quality impacts.
- Developed an internal **Python library** in collaboration with water research scientists to perform scientific and statistical analysis on diverse water quality and management time-series datasets.
- Developed python scripts to be scheduled as production jobs and workflows to automate water sampling schedules and water quality results to over 1,000 water systems, eliminating manual reporting.
- Implemented a **RabbitMQ**-based pub-sub system to enhance microservice performance and responsiveness by reducing synchronous task dependencies, cutting latency by 50%.
- Leveraged **T-SQL** programming to create procedures, triggers, and database scripts within MS SQL for various data workflows.
- Configured bitbucket pipelines to test and deploy web applications enabling automated **CI/CD** with **Jenkins**.

Aaseya — **Hyderabad, India**
Software Engineer — **Aug 2020 – Jun 2022**

- Developed secure and scalable RESTful APIs using **Spring Boot** to facilitate the exchange of court data among various entities within a web-based court management system of the Ministry of Justice of Saudi Arabia.
- Designed and Implemented key software modules for court operations, including hearings, case registration, and administrative tools, using **PEGA**(browser-based web development tool), while also managing database schema design and stored procedures in MS SQL.

JK Tech — **Hyderabad, India**
PEGA Developer — **Jan 2019 – Aug 2020**

- Designed and developed web application using PEGA for the Education and Real Estate Regulatory Authority departments within *One Portal*, a large-scale web system for the Andhra Pradesh state government, enabling features like exam booking, school affiliations, and real estate application approvals.
- Integrated the CCAvenue **payment gateway** necessary in various modules using **Java** and PEGA, enabling transactions for exam scheduling and regulatory fee payments.

Virtusa — **Hyderabad, India**
Associate Software Engineer — **Aug 2017 – Jan 2019**

- Implemented a multi-stage claim processing workflow in PEGA for AIG involving database integration and APIs for efficient data exchange and amongst various entities within the system.

SKILL SET

Languages: Python, Java, C, C++ **Frameworks:** React, Flask, Angular, SQLAlchemy, FastAPI, Spring Boot, Pandas, pyTorch, LangChain
Databases: MS SQL, MySQL, MongoDB **Tools/Softwares:** Git, PEGA (v7.6 – 8.0), VS Code, Maven, Docker, Kubernetes, AWS, Jenkins

EDUCATION

California State University, Sacramento	Master of Science, Computer Science	Aug 2022 – Dec 2024
Jawaharlal Nehru Technological University, India	Bachelor of Technology, Information technology	Aug 2013 – May 2017

PROJECTS

Hash-based Proof of Stake Consensus Protocol

- Researched and implemented a novel proof-of-stake consensus algorithm to address centralization issues in traditional protocols, simulating a blockchain network with 100 nodes using Python, Sockets, Multithreading and RSA encryption.
- Achieved a 15-20% improvement in decentralization metrics, such as Fairness and Entropy, enhancing the efficiency and integrity of transactions in the network.

Social Media Application with Enhanced Security

- Developed a social media web app with React and Spring Boot and MongoDB, using Diffie-Hellman asymmetric encryption to secure post and comment data by separating metadata and actual content across two servers.
- Used **AWS Elastic Container Service(ECS),** Auto-Scaling Groups and **Docker** for containerized deployment and scale management.

Brain Tumor Detection using Deep Learning

- Developed a web application using React and Python to classify three types of brain tumors with T1-Weighted MRI scans, using image classification techniques like **Vision Transformers, Fast-RCNN, and YOLO.**

SpotifAI

- Developed a web application with Spotify OAuth authentication and Spotify Web API that enables users to have insights into their playlists through a chatbot using LLM and **Langchain Agents**.